Genese: execution architecture and measured performance

World resolves policy intents, commits state and events, and admits decision work within execution capacity.

Native execution capabilities

Paper Table S13 · 10 mechanisms × 6 frameworks

Capability	Genese	OneSim	OASIS	MiroFish	AgentSociety	MatrAIx
Array-based ECS storage	**Yes**	N/D	N/D	N/D	N/D	N/D
Synchronized state persistence	**Yes**	Yes	No	No	No	No
Native action-rule enforcement	**Yes**	Partial	Yes	Yes	Yes	Partial
Policy-intent / world-effect separation	**Yes**	No	No	No	No	No
Atomic state–event commit	**Yes**	No	No	No	No	No
World-step isolation / rollback	**Yes**	No	No	No	Partial	Partial
Recovery / resume	**Yes**	No	No	No	Yes	Yes
Execution / connection pooling	**Yes**	Yes	No	No	Yes	No
Task admission / flow control	**Yes**	Partial	Yes	Yes	Yes	Yes
Decision / trial concurrency control	**Yes**	Partial	Yes	Yes	Yes	Yes

Yes: implemented Partial: part of the defined mechanism No: absent in the inspected path N/D: not documented

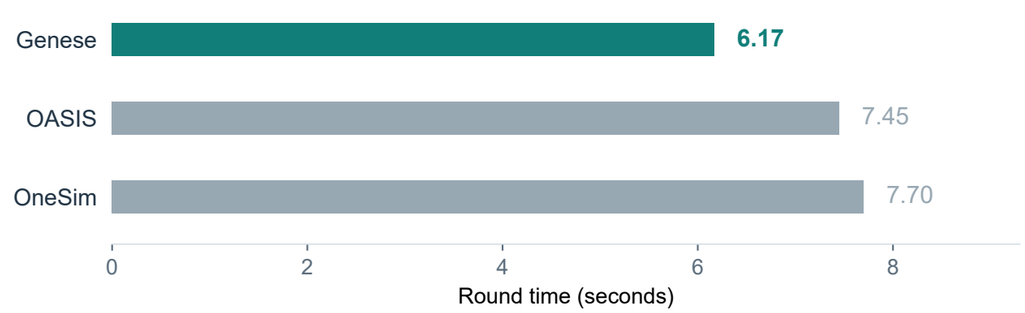
Action rules: OASIS/MiroFish platform rules; AgentSociety permissions and workspace. Resume: AgentSociety checkpoints; MatrAIx job progress.

OneSim: PostgreSQL supports synchronized persistence; event/interface and decision-flow controls are Partial. MatrAIx: questionnaire-output checks are Partial.

AgentSociety and MatrAIx isolate independent tasks; whole-world rollback is a separate mechanism.

Model execution · sharp-421m

1,000 agents · one T4 replica · two reported blocks

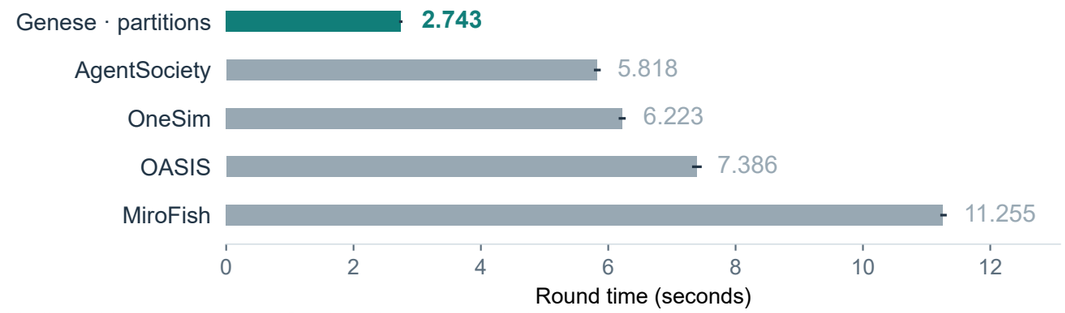


Every agent makes one accepted choice and writes back state.

MatrAIx parallel questionnaire adapter: 1,062.153 s on the immediate-backend workload.

Execution overhead · immediate backend

1,000 agents · 2×Xeon 8268 / 96 threads · 20 blocks



Medians; black marks show IQR. Initialization is excluded.